Question			Answer	Marks	AOs	Guidance	
4	(a)	(i)	Sequence is periodic [with period 4]	B1 [1]	1.2	Do not allow repeating, recurring etc.	The sequence can also be described as oscillating
4	(a)	(ii)	Total of 200 terms is $50\times(2+3+0+3)=400$	B1 [1]	1.1b	cao	
4	(b)	(i)	Sequence divergent for Either $b>1$ or $b\leq -1$	 B1 B1 [2]	 1.1b 1.1b	 Allow for one correct inequality Must have "or" or the union of sets Condone $b<-1$ or $\lvert b\rvert>1$	Note for $b=1$, the **sequence** is convergent, but the corresponding **series** is divergent
4	(b)	(ii)	Infinite sum of geometric series with $a=\frac{1}{3}, r=\frac{1}{3}$ $S=\frac{a}{1-r}=\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{1}{2}$	M1 A1 [2]	1.1b 1.1b	Using the sum of geometric series with $r=\frac{1}{3}$ www	